MILITARY SPECIFICATION SHEET

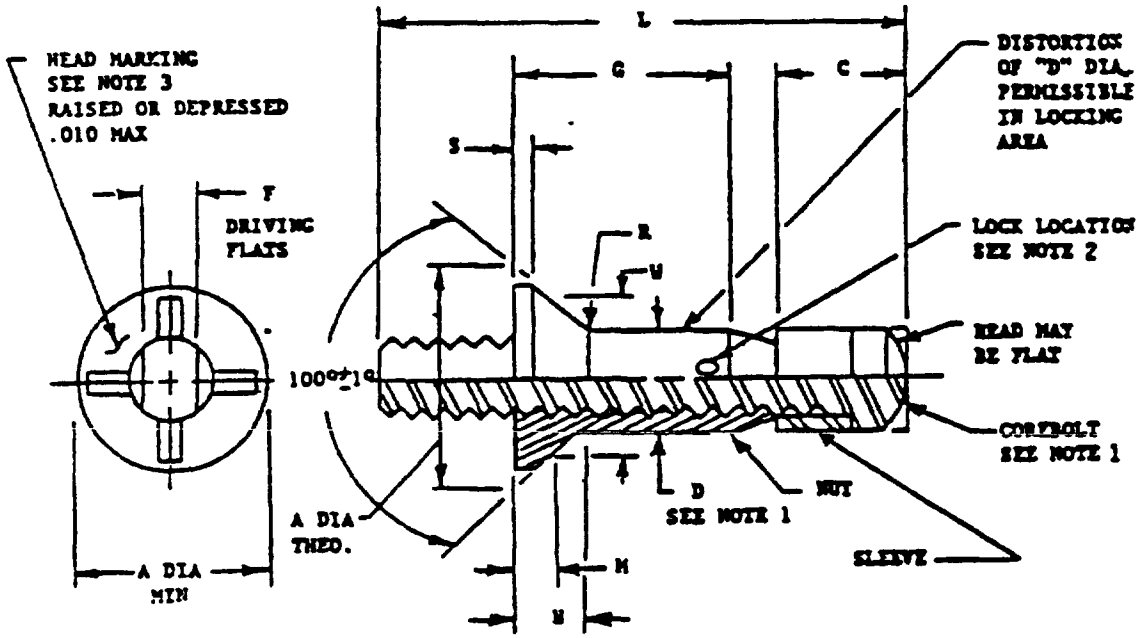
FASTENERS, BLIND, HIGH STRENGTH, INSTALLATION FORMED, THREADED, SELF-LOCKING, 100° FLUSH HEAD, 450°F, ALLOY STEEL 112 KSI Fsu

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the fasteners described herein shall consist of this document and the latest issue of specification MIL-F-81177.

Design and usage limitations - These blind fasteners shall be used in accordance with the limitations of MIL-F-81177/1.

Tooling for installation per MIL-F-81177-Appendix B.



NOTES:
1. Outside diameter of head of corebolt and outside diameter of sleeve shall be not greater than maximum "D" diameter.
2. Locking feature shall be visible in this area.
3. Head marking shall consist of the manufacturer's identification and the numerals "150."
4. Concentricity of nut head conical surface to "D" diameter shall be within .005 FIM (Full Indicator Movement).

FIGURE 1. Blind fastener.

FSC 5320

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:03Z
Check the source to verify that this is the current version before use.

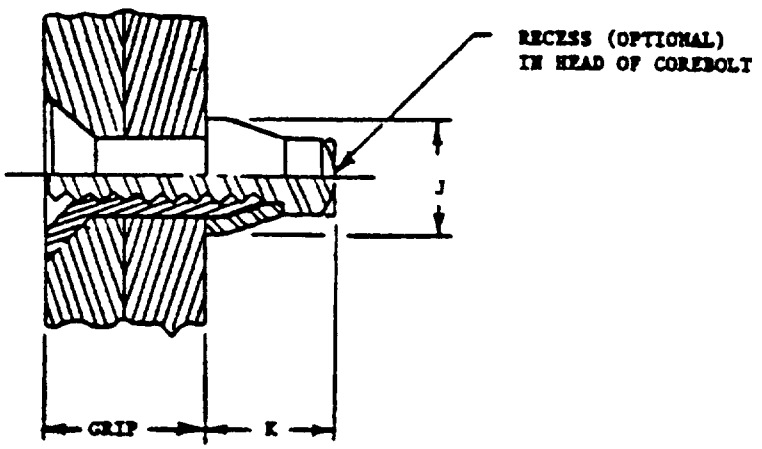


FIGURE 2. Typical installation.

REQUIREMENTS

Material:	Nut & Corebolt:	-5% Chrome Steel (H-11) per AMS 6487
	Sleeve:	-INCONEL X-750 per AMS5667, AMS5582 or A-286 Alloy per AMS 5731, AMS 5732 or AMS 5737
Heat treat:	Nut & Corebolt:	-Rockwell "C" 48-53
Protective Treatment:		
	Nut & Corebolt.	-Cadmium Plate per NAS672 with Supplementary Chromate treatment per QQ-P-416, color nut black
	Sleeve:	-Cadmium Plate per QQ-P-416, Type II, Class 2.
Lubricant:	Lubricant listed below may be applied to any or all components if required for performance.	
	Cetyl Alcohol per MIL-L-87132, Lauric Acid, WWM Carbo-Wax,	

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:03Z
Check the source to verify that this is the current version before use.

TABLE I Sizes and dimensions

DASH NO	NOM DIA	A DIA THEO	A DIA MIN	C MAX	D DIA	F DRIVING FLATS	H REF	J DIA MAX	K MAX	M GAGE PROT	R RAD	S MAX	W GAGE DIA	DOUBLE SHEAR STRENGTH LBS. MIN.	TENSILE STRENGTH LBS. MIN.
-5	1635	.332 325	.296	287	.1645 .1625	086 081	069	.244	.257	.0276 0238	.040 035	012	.2671 .2669	4,680	1,350
-6	1980	.385 .378	342	.326	.1990 1970	.104 .099	.077	.300	.288	.0250 .0190	.050 .045	.015	.3272 .3270	6,900	2,100
-8	2590	507 .499	463	.360	2600 .2580	.135 .130	.102	.384	.320	.0327 .0263	.060 .055	.015	.4320 .4318	11,800	3,650
-10	3105	.635 626	.577	440	3115 3095	.152 147	134	.427	.396	0390 .0320	.070 060	.020	.5451 .5449	17,500	5,200
-12	3735	.762 .752	.696	.539	.3745 .3725	.185 .180	.160	.516	473	.0452 .0378	.070 .060	.023	.6582 .6580	24,500	7,500

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:03Z
Check the source to verify that this is the current version before use.

TABLE II. Body diameters and grip lengths

SECOND DASH NUMBER	GRIP RANGE		G ±.020 -.000	L (Ref)				
	MIN	MAX		-5	-6	-8	-10	-12
-2	.094	.156	.156	.713	.829	-	-	-
-3	.157	.219	219	.775	.892	910	-	-
-4	.220	.281	.281	.838	.954	972	1.189	1.261
-5	282	.344	.344	.901	1.017	1.035	1.252	1 324
-6	345	406	.406	.963	1.079	1 097	1 314	1 386
-7	.407	.469	.469	1.026	1.142	1 160	1.377	1 499
-8	.470	.531	531	1 088	1.204	1 222	1.439	1.511
-9	.532	.594	.594	1.151	1.267	1 285	1.502	1 574
-10	.595	.656	.656	1.231	1.329	1 347	1 564	1 636
-11	.657	.719	.719	1 276	1.392	1 410	1 627	1 699
-12	.720	.781	.781	1 338	1 454	1.472	1 689	1 761
-13	782	.844	.844	1 401	1.517	1 535	1.752	1 824
-14	845	.906	.906	1 463	1.579	1 597	1 814	1 886
-15	.907	.969	.969	1 526	1 642	1.660	1.877	1 949
-16	970	1.031	1.031	1.588	1 704	1.722	1 939	2 011
-17	1.032	1.094	1.094		1 767	1.785	2 002	2 074
-18	1 095	1.156	1.156		1 829	1 847	2.064	2 135
-19	1.157	1.219	1.219		1.892	1.910	2.127	2 199
-20	1 220	1 281	1.281		1 954	1 972	2 189	2 261
-21	1 282	1 344	1.344		2 017	2 035	2.252	2 324
-22	1 345	1.406	1 406		2 079	2 907	2 314	2 386
-23	1 407	1.469	1 469		2.142	2 160	2 377	2 449
-24	1 470	1.531	1 531		2 204	2 222	2 439	2 511
-25	1.532	1.594	1.594		2 267	2 285	2.502	2 574
-26	1 595	1 656	1 656		2.329	2 347	2 564	2 636
-27	1 657	1 719	1 719		2.392	2 410	2 627	2 699
-28	1 720	1 781	1 781		2 454	2 472	2 698	2 761
-29	1 782	1 844	1 844		2 517	2 535	2 752	2.824
-30	1 845	1 906	1 906		2 579	2 597	2 814	2 885
-31	1 907	1 969	1 969		2 642	2 660	2 877	2 949
-32	1 970	2 031	2 031		2 704	2 722	2.939	3 011

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:03Z
Check the source to verify that this is the current version before use.

EXAMPLE OF PART NUMBER

M81177/2-XX-XX

XX-XX — Grip Lengths in 16ths (See Table II).

XX — Body Diameter in 32nds (See Table II).

M81177/2- — Basic Number

Custodians
Army - AV
Air Force - 11
Navy - AS

Review Activities
Air Force - 99
DLA - IS
Army - AR, MI

Preparing Activity
Navy - AS

Project No 5320-0308-2

U.S. G.P.O 703-023/4879

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:03Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T05:03Z
Check the source to verify that this is the current version before use.